From a reference image to specimen data acquisition

This workflow covers project definition and specimen data acquisition in SMORPHILA. A reference image defines landmark identities, ordered segment groups, angular constraints, and an analytical reference frame. The same protocol is then applied to individual specimens. In this example, angles within each polygon are retained while articulation angles between polygons are standardized.

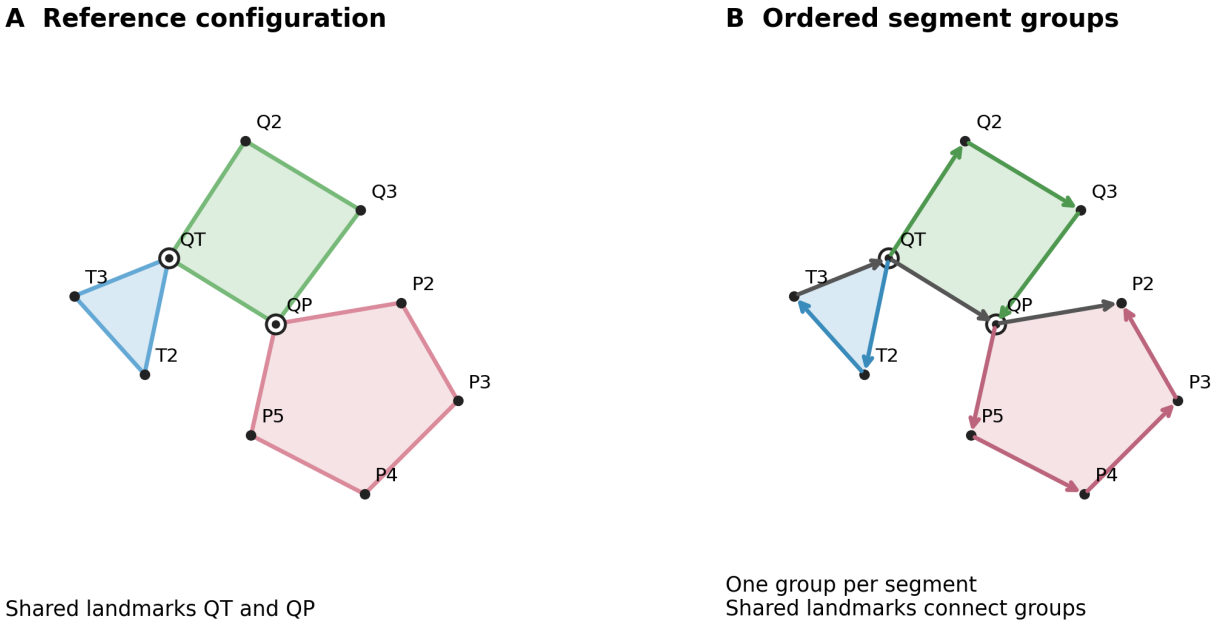


Figure 1. Definition of an articulated landmark configuration in SMORPHILA. (A) Landmarks are placed at the vertices of a triangle, a quadrilateral, and a pentagon. Shared landmarks QT and QP connect adjacent polygons; the polygons need not be regular. (B) Each segment is assigned to one ordered group, while landmarks may be shared. The three polygon groups are distinguished by colour, and the connecting group is shown in grey. In this example, Keep current angle retains the angles within each polygon, while prescribed articulation angles control their relative orientation. Both panels show the reference configuration before posture standardization. The resulting idealized configuration and analytical reference frame are shown in Figure 2.

Defining the configuration

A Define an articulated configuration

Start from a reference image containing three connected polygonal structures: a triangle, a quadrilateral, and a pentagon. Place landmarks at their vertices. Shared landmarks connect the polygons and define the articulated structure.

B Preserve polygon shape and change articulation angles

Create ordered segment groups for the polygonal structures. Each segment should belong to one group only, whereas landmarks may be shared between groups. In this example, a separate connecting group contains T3–QT, QT–QP, and QP–P2; the remaining boundary segments belong to the three polygon groups.

In this example, select Keep current angle for the angles within each polygon. These angles are measured in the reference image and retained during subsequent rotations. Prescribe new values only for the articulation angles that control the relative orientation of the polygons. Each polygon therefore rotates as a rigid unit about its shared landmark, preserving its internal angles and side lengths.

Panel caption Group definition and constraints. Angles within each polygon are retained from the reference image. Only articulation angles between polygons are changed; polygon shapes, segment lengths, and connectivity are preserved.

C Set the analytical reference frame

Select an origin landmark and define an axis using two landmarks. In this example, QP defines the origin and QP–P2 defines the axis. After setting the articulation angles, SMORPHILA rotates the entire idealized configuration into the requested orientation while keeping QP fixed. This final alignment changes neither the internal polygon angles nor the newly prescribed articulation angles. Applying the same settings provides a common reference-frame convention for all specimens.

Panel caption Analytical reference frame. The entire idealized configuration is aligned to a common axis by a rigid rotation about the selected origin. Polygon shapes and the prescribed articulation angles remain unchanged.

The idealized configuration in SMORPHILA

The editor view below shows the resulting configuration and the segment-angle annotations. The distinction is between angles within a polygon, which are retained, and articulation angles between polygons, which are prescribed to standardize posture.



Figure 2. Idealized configuration displayed in the SMORPHILA Landmark Definition Editor. Blue arrows indicate ordered segments, red points indicate landmarks, and yellow annotations report segment orientations or turning angles. The angles within the triangle, quadrilateral, and pentagon are retained from the reference image. Only the articulation angles are changed, producing the aligned connecting chain T3–QT–QP–P2. QP is the selected origin, and QP–P2 defines the analytical axis.

An absolute angle specifies a segment orientation, whereas a turning angle specifies the change in direction between consecutive ordered segments. These editor labels should therefore be read in relation to the segment groups rather than interpreted as polygon interior angles.

Applying the project to individual specimens

D Acquire homologous landmarks

In the data acquisition module, identify the corresponding polygon vertices in each specimen image using the landmark names defined in the project. Digitize each shared articulation landmark once, so that the connected groups use the same point. These acquired coordinates describe the specimen in its original posture.

E Inspect the standardized configuration

Apply the project definitions to the acquired landmarks. The standardized configuration uses the specimen's segment lengths and the prescribed articulation settings. In this example, within-polygon angles are retained so that each polygon keeps its shape, while its orientation relative to the other polygons changes. The analytical reference frame controls the orientation of the complete configuration.

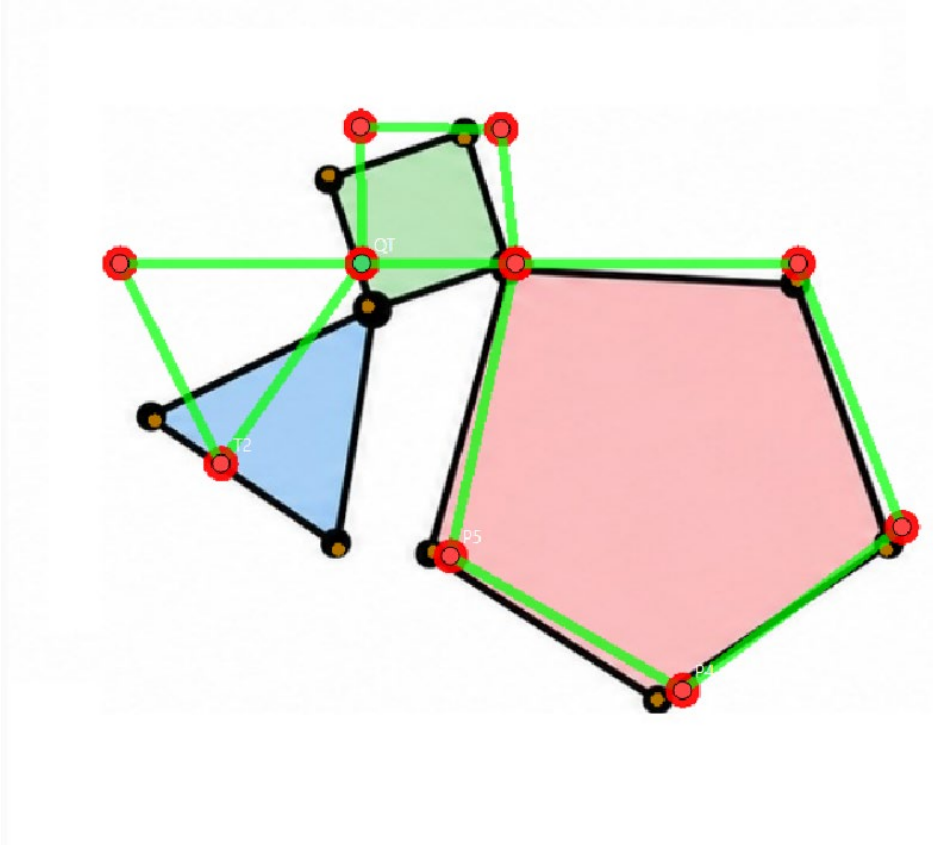


Figure 3. Data acquisition and standardization for a specimen with a relatively large pentagon. The image outlines show the original polygon configuration. Ochre markers indicate acquired landmarks; green segments and red markers show the standardized configuration overlaid on the image. The overlay allows the user to inspect how the project's articulation settings reposition the polygons while retaining their segment lengths and within-polygon geometry.

Applying the same protocol across specimens

A second specimen illustrates variation in relative polygon size and initial posture. Landmark identities, group connectivity, angular constraints, and reference-frame settings remain the same. The resulting configuration retains the specimen's dimensions while expressing the articulation prescribed by the project.

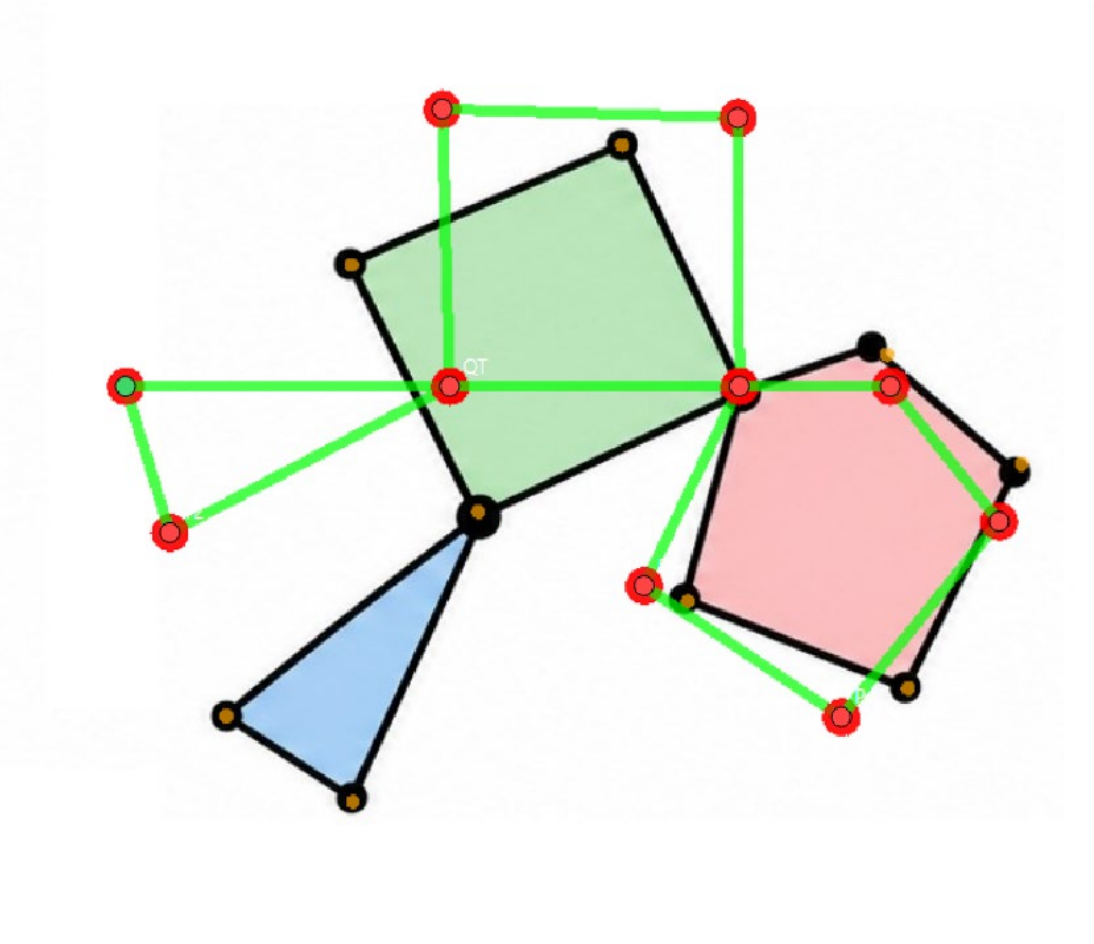


Figure 4. The same standardization protocol applied to a second specimen with different polygon proportions and articulation angles. Ochre markers indicate acquired landmarks on the original image; green segments and red markers represent the standardized configuration. Comparison with Figure 3 illustrates the application of shared project definitions to specimens that differ in size and posture.

From acquired to analytical coordinates

The acquired landmarks record the original posture, whereas the standardized landmarks provide the configuration prepared for subsequent morphometric analyses. Reviewing both configurations together helps assess landmark placement and the effects of the angular constraints before using the resulting coordinate data.